

§ 1.4.4 电势的计算

一、电势叠加原理

$$\varphi = \int_{(P)}^{(\text{电势零点})} \vec{E} \cdot d\vec{l} = \int_{(P)}^{(\text{电势零点})} \sum_i \vec{E}_i \cdot d\vec{l} = \sum_i \int_{(P)}^{(\text{电势零点})} \vec{E}_i \cdot d\vec{l}$$

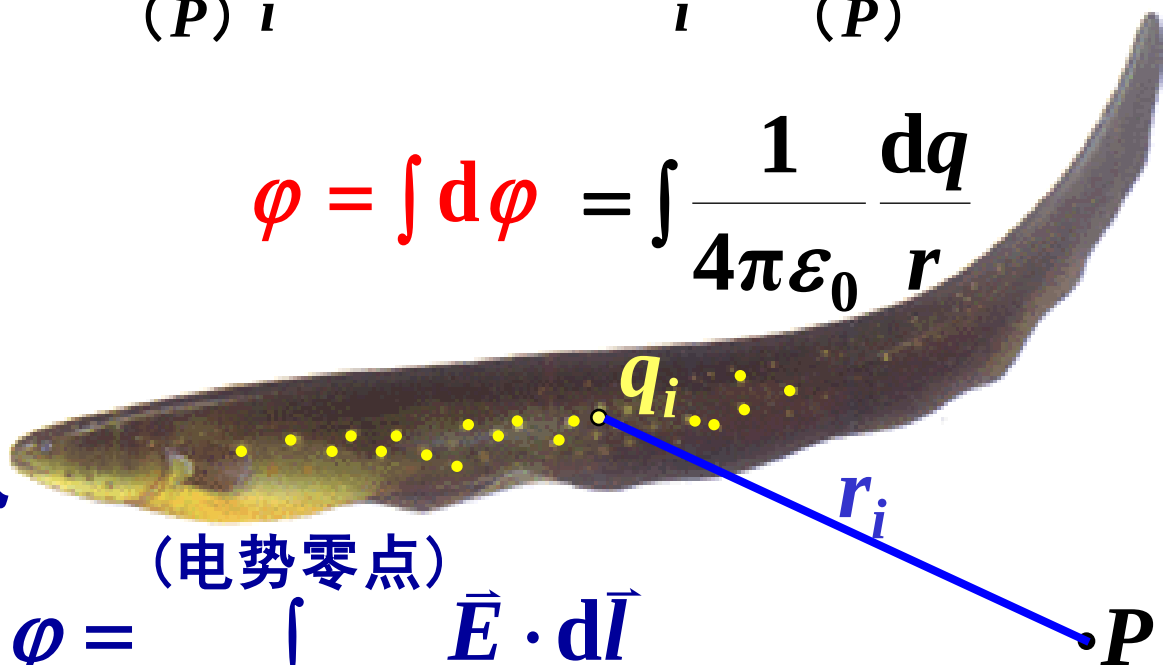
$$\varphi = \sum_i \varphi_i = \sum_i \frac{q_i}{4\pi\epsilon_0 r_i}$$

$$\varphi = \int d\varphi = \int \frac{1}{4\pi\epsilon_0} \frac{dq}{r}$$

二、任意带电体电势

1) 由定义式出发 $\varphi = \int_{(P)}^{(\text{电势零点})} \vec{E} \cdot d\vec{l}$

2) 电势叠加原理 $\varphi = \sum_i \varphi_i$ $\varphi = \int d\varphi$

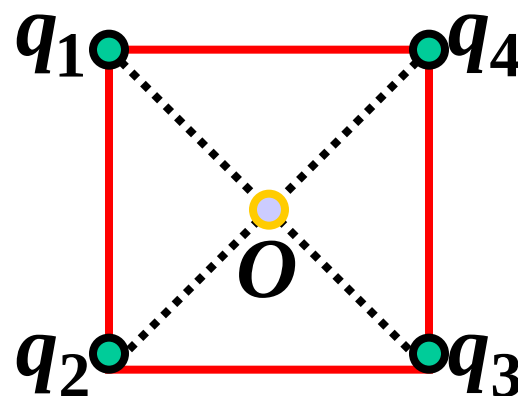


[例] 点电荷 q_1, q_2, q_3, q_4 均为 $4.0 \times 10^{-9} \text{ C}$, 放置在一正方形的四个顶角上, 各顶角与正方形中心 O 的距离均为 5.0 cm , (1) 计算 O 点的电势; (2) 将试验电荷 $q_0 = 1.0 \times 10^{-9} \text{ C}$ 从无限远处移到 O 点, 电场力做的功为多少? (3) 电势能改变多少? 是增加还是减少?

解: (1) 选无限远处为电势零点, 根据电势叠加原理, O 点电势为:

$$\varphi_O = \varphi_1 + \varphi_2 + \varphi_3 + \varphi_4$$

$$= 4\varphi_1 = 4 \frac{q_1}{4\pi\epsilon_0 r} = 2880 \text{ V}$$



$$(2) A = q_0(\varphi_\infty - \varphi_O) = -2.88 \times 10^{-6} \text{ J}$$

$$(3) W_\infty - W_O = A = -2.88 \times 10^{-6} \text{ J}$$

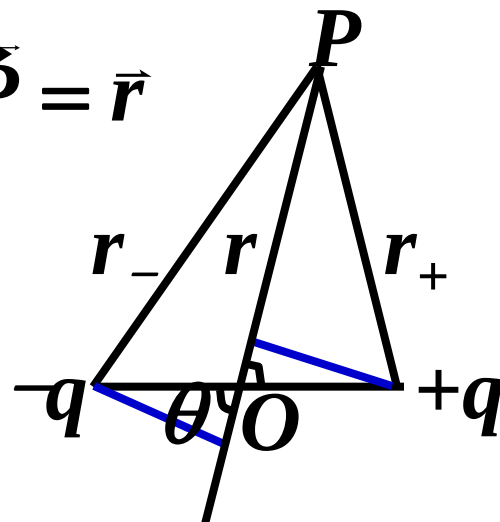
电势能改变 $W_O - W_\infty = 2.88 \times 10^{-6} \text{ J}$ 增加

[例] 求距电偶极子相当远处的电势。

解:

$$\varphi_+ = \frac{1}{4\pi\epsilon_0} \frac{q}{r_+} \quad \varphi_- = \frac{1}{4\pi\epsilon_0} \frac{(-q)}{r_-}$$

$$\overrightarrow{OP} = \vec{r}$$



$$\varphi = \varphi_+ + \varphi_- = \frac{1}{4\pi\epsilon_0} \frac{q}{r_+} + \frac{1}{4\pi\epsilon_0} \frac{(-q)}{r_-}$$

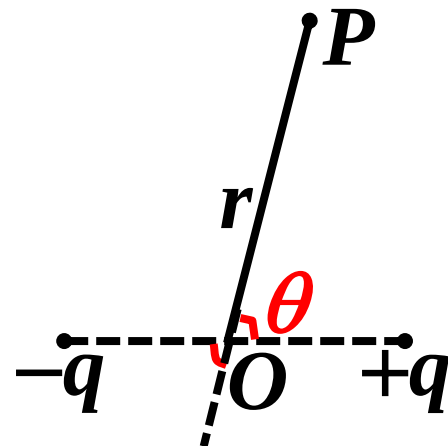
$$r_+ \approx r - \frac{l}{2} \cos \theta \quad r_- \approx r + \frac{l}{2} \cos \theta$$

$$\varphi = \frac{1}{4\pi\epsilon_0} \frac{q}{r - \frac{l}{2} \cos \theta} - \frac{1}{4\pi\epsilon_0} \frac{q}{r + \frac{l}{2} \cos \theta} \approx \frac{1}{4\pi\epsilon_0} \frac{ql \cos \theta}{r^2}$$

$$= \frac{1}{4\pi\epsilon_0} \frac{\vec{p} \cdot \hat{e}_r}{r^2} = \frac{1}{4\pi\epsilon_0} \frac{\vec{p} \cdot \vec{r}}{r^3}$$

[例] 求距电偶极子相当远处的电势。

解： $\varphi \approx \frac{1}{4\pi\epsilon_0} \frac{ql \cos \theta}{r^2} = \frac{1}{4\pi\epsilon_0} \frac{\vec{p} \cdot \hat{e}_r}{r^2}$



1. 延长线上, $\theta = 0$:

$$\varphi = \frac{1}{4\pi\epsilon_0} \frac{p}{r^2}$$

$$\theta = \pi: \quad \varphi = -\frac{1}{4\pi\epsilon_0} \frac{p}{r^2}$$

2. 中垂面上, $\theta = \pi/2$: $\varphi = 0$

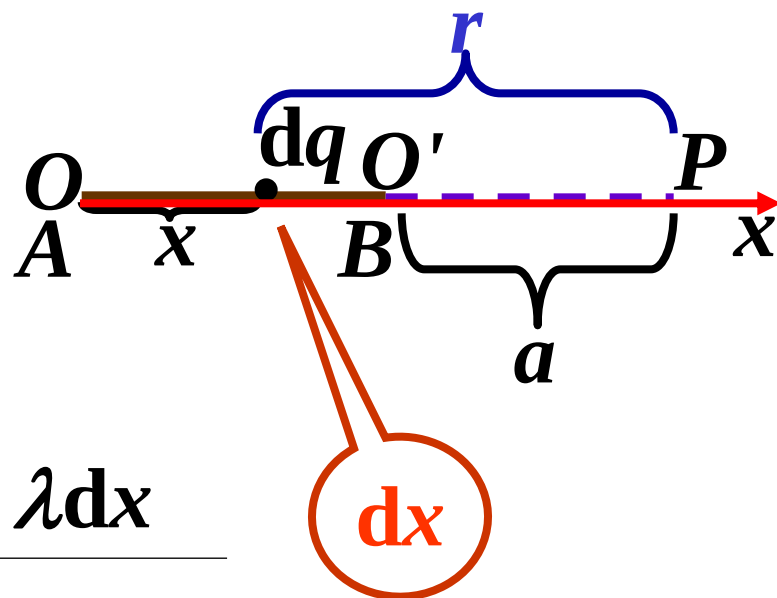
[例] 均匀带电直线，长 l ，带电为 Q ，求其延长线上离杆 B 端距离为 a 的 P 点的电势。

解： 取 dq

$$dq = \lambda dx = \frac{Q}{l} dx$$

$$d\varphi_P = \frac{1}{4\pi\epsilon_0} \frac{dq}{r} = \frac{1}{4\pi\epsilon_0} \frac{\lambda dx}{l + a - x}$$

$$P \text{ 点电势 } \varphi_P = \int_0^l \frac{\lambda}{4\pi\epsilon_0} \frac{dx}{l + a - x} = \frac{1}{4\pi\epsilon_0} \frac{Q}{l} \ln \frac{l + a}{a}$$

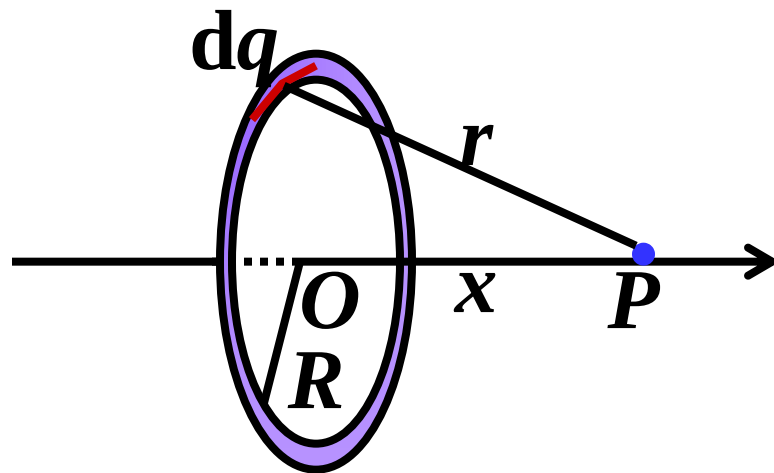


[例] 求均匀带电圆环轴线上的电势 (R, q)。

解: $d\varphi = \frac{1}{4\pi\epsilon_0} \frac{dq}{r}$

$$\varphi = \int d\varphi = \int \frac{1}{4\pi\epsilon_0} \frac{dq}{r}$$

$$= \frac{1}{4\pi\epsilon_0} \frac{q}{r} = \frac{1}{4\pi\epsilon_0} \frac{q}{\sqrt{R^2 + x^2}}$$



讨论

1. $x \gg R$ $\varphi = \frac{1}{4\pi\epsilon_0} \frac{q}{x}$

点电荷电势

2. $x = 0$ $\varphi = \frac{1}{4\pi\epsilon_0} \frac{q}{R}$

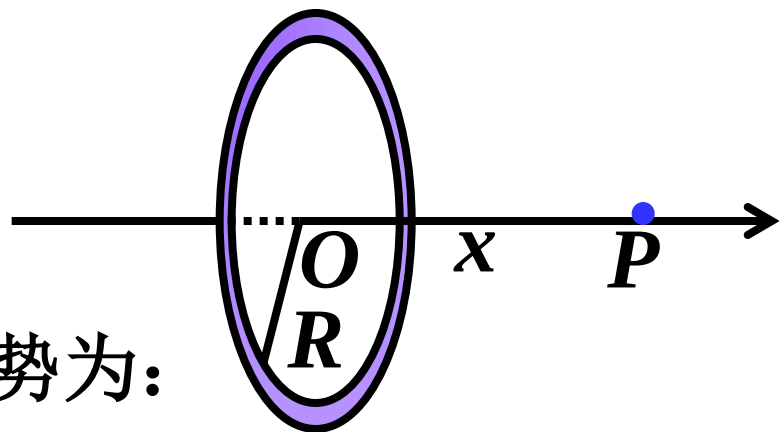
此处
 $E = 0$

3. 电量可非均匀分布。

[例] 求均匀带电圆环轴线上的电势 (R, q)。

另解： 圆环轴线上任一点的场强为：

$$\vec{E} = \frac{qx}{4\pi\epsilon_0(x^2 + R^2)^{3/2}} \hat{i} \quad \text{方向沿轴线指向无限远}$$



距环心为 x 处 P 点的电势为：

$$\begin{aligned} \varphi &= \int_P^\infty \vec{E} \cdot d\vec{l} = \int_P^\infty E dx \\ &= \int_x^\infty \frac{qx}{4\pi\epsilon_0(x^2 + R^2)^{3/2}} dx \\ &= \frac{q}{4\pi\epsilon_0 \sqrt{x^2 + R^2}} \end{aligned}$$

[例] 已知：均匀带电圆盘，总电量 Q ，半径 R 。

求：圆盘轴线上任意点 P 处的电势。

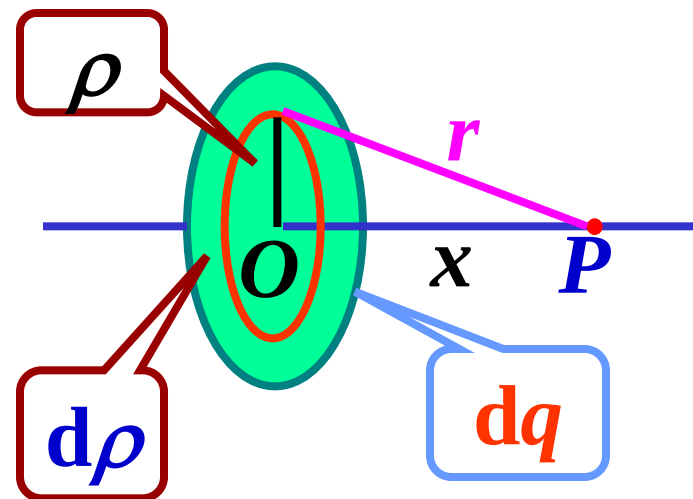
解：利用电势积分法。

1. 分割合适电荷元

$$dq = \sigma dS$$

$$\sigma = \frac{Q}{\pi R^2}$$

$$dS = 2\pi \rho d\rho$$



2. 电荷元 dq 在 P 处电势

$$d\varphi = \frac{dq}{4\pi\epsilon_0 r}$$

3. 总电量 Q 在 P 处电势

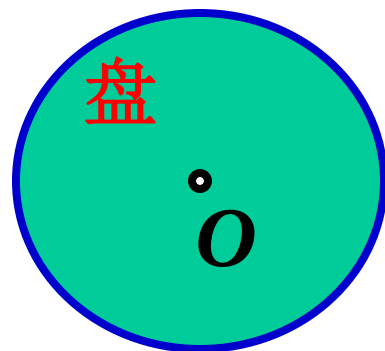
$$\varphi = \int_Q d\varphi = \int_Q \frac{dq}{4\pi\epsilon_0 (\rho^2 + x^2)^{1/2}}$$

$$\varphi = \int_Q \mathbf{d}\varphi = \int_Q \frac{\mathbf{d}q}{4\pi\epsilon_0(\rho^2 + x^2)^{1/2}}$$



当 $x = 0$ 时,

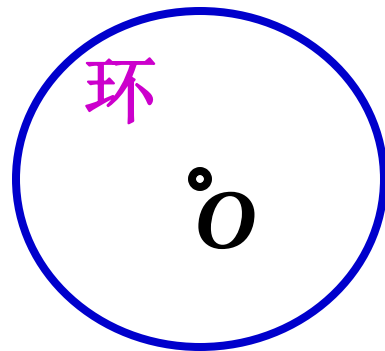
$$\varphi_{\text{盘}0} = \frac{Q}{2\pi\epsilon_0 R}$$



盘心处电势不为零。

$$\begin{aligned} \varphi &= \int_0^R \frac{\frac{Q}{\pi R^2} 2\pi \rho \mathbf{d}\rho}{4\pi\epsilon_0(\rho^2 + x^2)^{1/2}} \\ &= \frac{Q}{4\pi\epsilon_0 R^2} 2\sqrt{\rho^2 + x^2} \Big|_0^R \\ &= \frac{Q}{2\pi\epsilon_0 R^2} (\sqrt{R^2 + x^2} - x) \end{aligned}$$

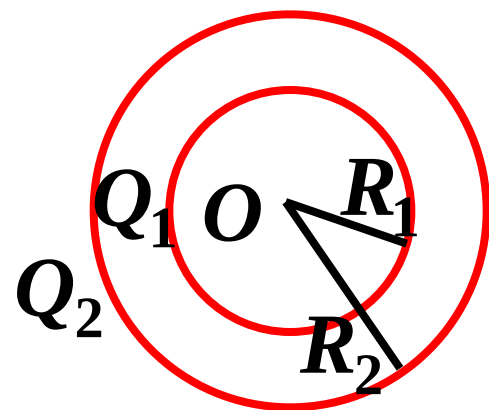
$$\varphi_{\text{环}0} = \frac{Q}{4\pi\epsilon_0 R}$$



注意: $\varphi_{\text{盘}0} = 2\varphi_{\text{环}0}$

[例] 两个半径分别为 R_1 和 R_2 的带电球面同心放置，所带电量分别为 Q_1 和 Q_2 ，求其电场的电势分布。

解：

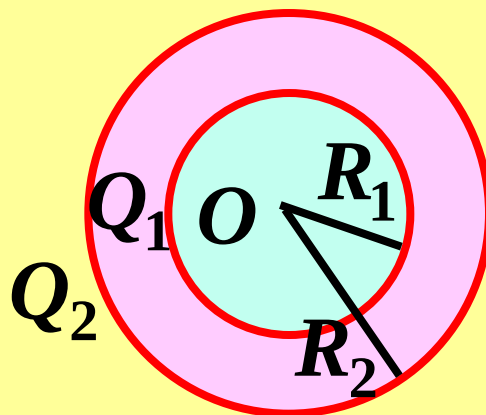


Q_1 产生的电势分布为

$$\varphi_1 = \begin{cases} \frac{Q_1}{4\pi\epsilon_0 R_1} (r \leq R_1) \\ \frac{Q_1}{4\pi\epsilon_0 r} (r > R_1) \end{cases}$$

Q_2 产生的电势分布为

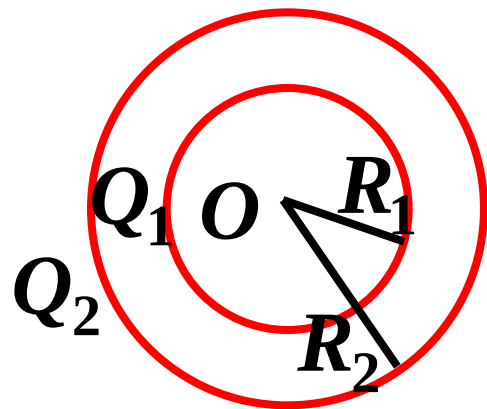
$$\varphi_2 = \begin{cases} \frac{Q_2}{4\pi\epsilon_0 R_2} (r \leq R_2) \\ \frac{Q_2}{4\pi\epsilon_0 r} (r > R_2) \end{cases}$$



区域	内壳贡献	外壳贡献	总电势
$r < R_1$	$\varphi_{1内} = \frac{Q_1}{4\pi\epsilon_0 R_1}$	$\varphi_{2内} = \frac{Q_2}{4\pi\epsilon_0 R_2}$	$\varphi_{1内} + \varphi_{2内}$
$R_1 < r < R_2$	$\varphi_{1外} = \frac{Q_1}{4\pi\epsilon_0 r}$		$\varphi_{1外} + \varphi_{2内}$
$r > R_2$		$\varphi_{2外} = \frac{Q_2}{4\pi\epsilon_0 r}$	$\varphi_{1外} + \varphi_{2外}$

解：总电势分布为

$$\varphi = \varphi_1 + \varphi_2$$



$$= \begin{cases} \frac{Q_1}{4\pi\epsilon_0 R_1} + \frac{Q_2}{4\pi\epsilon_0 R_2} & (r \leq R_1) \\ \frac{Q_1}{4\pi\epsilon_0 r} + \frac{Q_2}{4\pi\epsilon_0 R_2} & (R_1 < r \leq R_2) \\ \frac{Q_1 + Q_2}{4\pi\epsilon_0 r} & (r > R_2) \end{cases}$$

[例] 均匀带电球层 ρ , 内外半径为 R_1, R_2 , 求: A 、 B 两点的电势

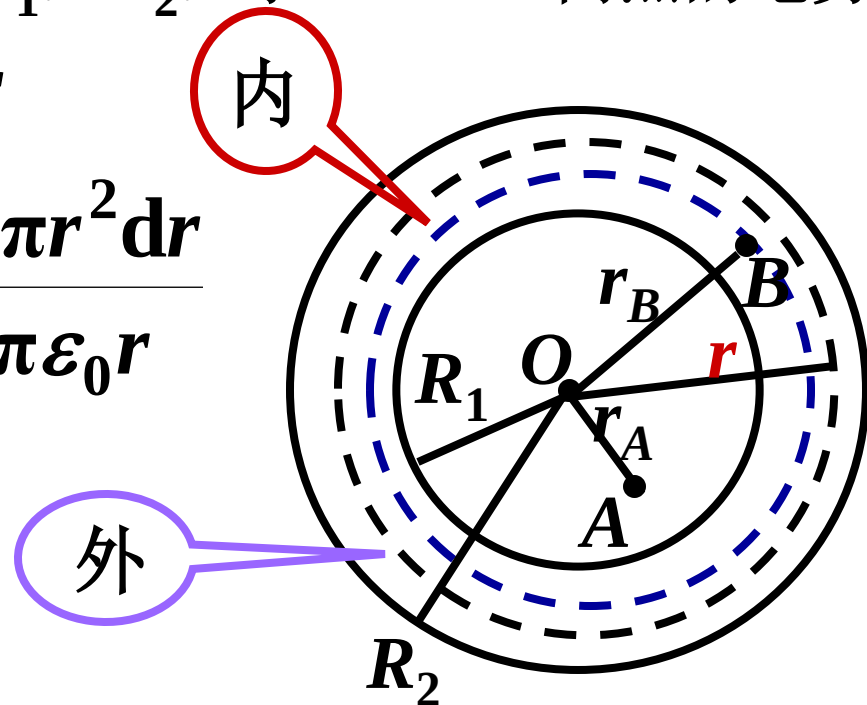
解: 取带电球面 $\mathrm{d}q = \rho 4\pi r^2 \mathrm{d}r$

1. 求 φ_A

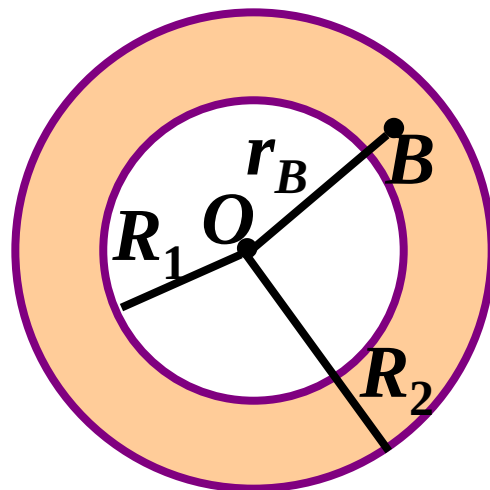
$$\begin{aligned}\varphi_A &= \int \mathrm{d}\varphi = \int_{R_1}^{R_2} \frac{\mathrm{d}q}{4\pi\epsilon_0 r} = \int_{R_1}^{R_2} \frac{\rho 4\pi r^2 \mathrm{d}r}{4\pi\epsilon_0 r} \\ &= \frac{\rho}{2\epsilon_0} (R_2^2 - R_1^2)\end{aligned}$$

2. 求 φ_B

$$\begin{aligned}\varphi_B &= \varphi_{B\text{外}} + \varphi_{B\text{内}} = \int_{r_B}^{R_2} \frac{\mathrm{d}q}{4\pi\epsilon_0 r} + \frac{q_{\text{内}}}{4\pi\epsilon_0 r_B} \\ &= \frac{\rho}{2\epsilon_0} (R_2^2 - r_B^2) + \frac{\rho}{4\pi\epsilon_0 r_B} \frac{4}{3} \pi (r_B^3 - R_1^3) \\ &= \frac{\rho}{6\epsilon_0} \left(3R_2^2 - r_B^2 - \frac{2R_1^3}{r_B} \right)\end{aligned}$$

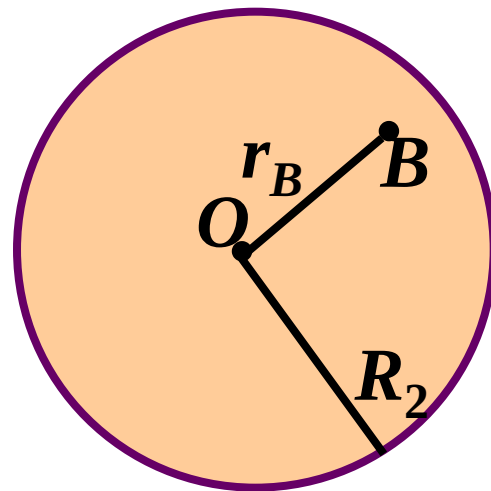


$$\varphi_B = \frac{\rho}{6\varepsilon_0} \left(3R_2^2 - r_B^2 - \frac{2R_1^3}{r_B} \right)$$



当 $R_1 = 0$ 时，均匀带电球体内

$$\varphi_B = \frac{\rho}{6\varepsilon_0} (3R_2^2 - r_B^2)$$



电势的计算方法

(一) 已知电荷分布求电势

1. 点电荷

$$\varphi = \frac{q}{4\pi\epsilon_0 r}$$

2. 点电荷系

$$\varphi = \Sigma \varphi_i = \sum_{i=1}^n \frac{q_i}{4\pi\epsilon_0 r_i}$$

3. 电荷连续分布的带电体

$$\varphi = \int d\varphi = \int \frac{dq}{4\pi\epsilon_0 r}$$

(二) 已知电场分布求电势

$$\varphi = \int_{P(\text{任意路径 } L)}^{\text{电势零点}} \vec{E} \cdot d\vec{l}$$

注意

(1) 零点的选取

(2) 最方便的路径

(3) E 分区时, 分区积分